

HUMAN PHYSIOLOGY - I

Unit II Notes: Blood, Lymphatic System and Immunity

Syllabus coverage: composition and functions of blood; cellular components and their functions; haemoglobin and anaemia; blood groups and coagulation; composition and function of lymph; lymphatic tissue; immunity with the role of thymus.

High-yield map	What to master
Blood	Definition, physical properties, composition, plasma proteins, serum, functions and formed elements.
RBC and Hb	RBC morphology, normal count, lifespan, fate, erythropoiesis, hemoglobin structure, synthesis, functions and iron metabolism.
Anaemia	Definition, morphological and etiological classification, symptoms, blood indices and common exam examples.
WBC and immunity	Types of WBC, differential count, properties, functions, innate and acquired immunity.
Platelets and coagulation	Platelet functions, hemostasis, clotting factors, intrinsic and extrinsic pathways, clot retraction and fibrinolysis.
Blood groups and lymph	ABO, Rh, cross-matching, transfusion reactions, lymph composition, lymph nodes, lymphatic tissues and thymus.

Presentation tip: For a 5-mark short note, write definition first, then 4-5 headings with values. For a 15-mark answer, add a labelled diagram, normal values, a table, and one applied/clinical point.

1. Blood - Definition, Properties and Composition

Blood is a connective tissue in fluid form. It is commonly described as the fluid of life because it transports oxygen from lungs to tissues and carbon dioxide from tissues to lungs. It is also called the fluid of growth because it carries absorbed nutrients and hormones, and the fluid of health because it removes waste products and protects the body against infection.

The average blood volume in a normal adult is about 5 L. It is approximately 8% of body weight in a healthy young adult. In adult females the volume is slightly lower, about 4.5 L. Blood is slightly alkaline with a pH of about 7.4. Arterial blood appears scarlet red because it contains more oxygen, whereas venous blood appears purple-red because it contains more carbon dioxide.

Property	Normal / important point
Colour	Arterial blood is scarlet red; venous blood is purple-red.
Volume	Adult: about 5 L; female: about 4.5 L; newborn: about 450 mL.
pH	Slightly alkaline: about 7.4.
Specific gravity	Total blood: 1.052-1.061; cells: 1.092-1.101; plasma: 1.022-1.026.
Viscosity	About five times more viscous than water, mainly due to RBCs and plasma proteins.

Compositionally, blood has two major parts: formed elements and plasma. Formed elements include red blood cells, white blood cells and platelets. Plasma is the straw-coloured liquid portion and forms about 55% of the blood, while RBCs form about 45%. The thin whitish layer between plasma and RBCs after centrifugation is the buffy coat, formed by WBCs and platelets.

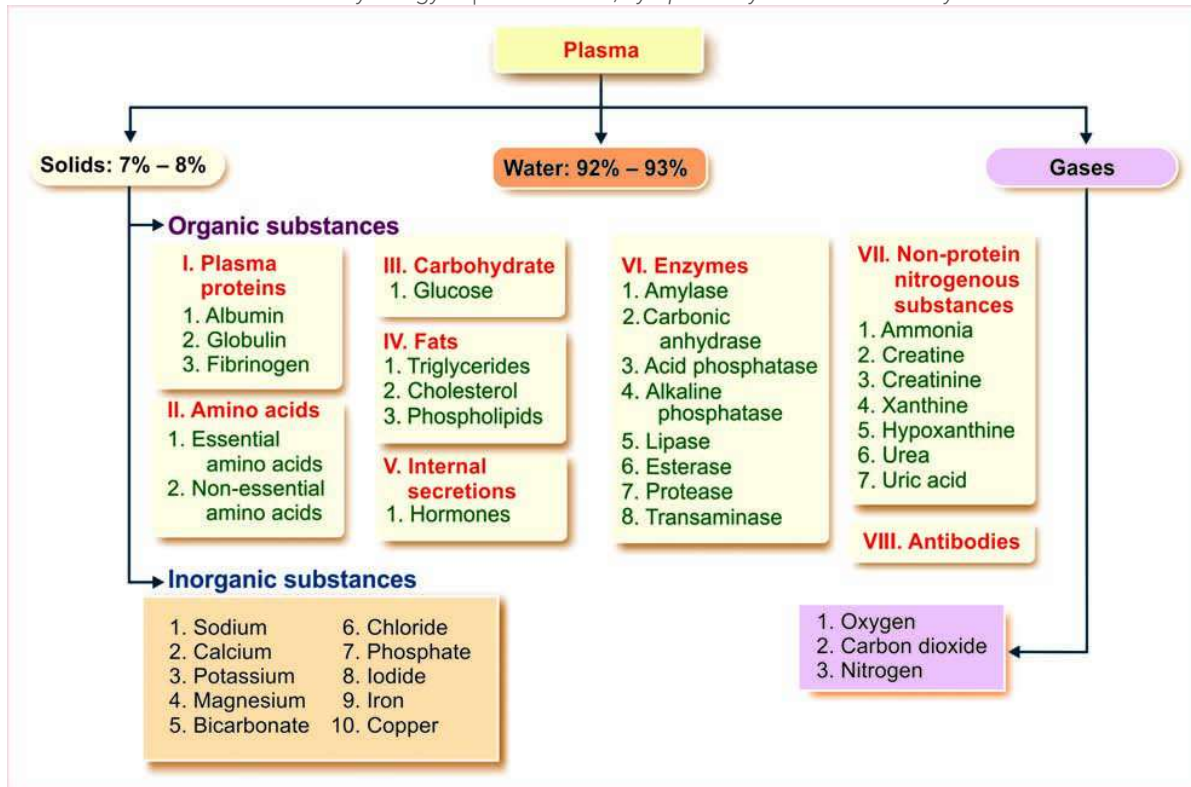


Figure: Composition of plasma and major plasma constituents. Source: Sembulingam.

Diagram tip: In composition questions, draw blood = plasma + cells first, then show plasma = water + organic + inorganic substances.

2. Plasma, Serum and Plasma Proteins

Plasma is the liquid portion of blood. It contains about 91-92% water and 8-9% solids. The solids include organic substances such as plasma proteins, glucose, amino acids, lipids, enzymes, hormones, urea and uric acid, and inorganic substances such as sodium, potassium, calcium, magnesium, chloride, bicarbonate and phosphate ions.

Serum is the clear straw-coloured fluid that separates from a blood clot. It is almost similar to plasma but does not contain fibrinogen and most clotting factors because these are used in clot formation. Therefore, plasma is used when clotting factors must be studied, while serum is commonly used for many biochemical investigations.

Feature	Plasma	Serum
Definition	Liquid part of anticoagulated blood.	Fluid that oozes from clotted blood.
Fibrinogen	Present.	Absent.
Clotting factors	Mostly present.	Absent or markedly reduced.
Use	Coagulation studies and plasma transfusion.	Routine biochemical and serological tests.
Appearance	Straw-coloured liquid.	Clear straw-coloured fluid.

Plasma proteins include albumin, globulins, fibrinogen and prothrombin. Albumin is the most important protein for maintaining colloid osmotic pressure, so it helps retain water inside blood vessels. Globulins include alpha, beta and gamma globulins; gamma globulins are immunoglobulins and have defensive functions. Fibrinogen and prothrombin are essential for clotting.

Plasma protein	Major functions
Albumin	Maintains colloid osmotic pressure; transports fatty acids, bilirubin, hormones and drugs; acts as a protein reserve.
Globulins	Transport substances; alpha and beta globulins transport lipids and metals; gamma globulins act as antibodies.

Fibrinogen	Converted into fibrin during coagulation and helps in clot formation.
Prothrombin	Precursor of thrombin; essential in the coagulation cascade.

Exam trap: Serum is not plasma without cells; serum is plasma without fibrinogen and most clotting factors. Write this clearly to avoid losing marks.

3. Functions of Blood

Blood performs transport, regulatory, protective and storage functions. These functions are not independent; for example, plasma proteins contribute to transport and buffering, RBCs transport gases and participate in acid-base balance, while WBCs and platelets protect the body by immunity and hemostasis.

Function	Explanation
Nutritive	Carries glucose, amino acids, lipids, vitamins and minerals from the digestive system to tissues.
Respiratory	Transports oxygen from lungs to tissues mainly by hemoglobin and carries carbon dioxide from tissues to lungs.
Excretory	Transports metabolic wastes such as urea, uric acid, creatinine and bilirubin to kidney, skin, liver and other excretory organs.
Hormonal transport	Carries hormones from endocrine glands to target tissues and also transports enzymes.
Water balance	Water in blood exchanges freely with interstitial fluid and helps maintain body fluid balance.
Acid-base balance	Hemoglobin and plasma proteins act as buffers and help maintain pH.
Temperature regulation	High specific heat of blood distributes heat and helps thermoregulation.
Storage	Acts as a reserve of water, proteins, glucose and electrolytes during starvation, fluid loss or electrolyte imbalance.
Defense	WBCs fight infection, lymphocytes produce immunity, and platelets prevent blood loss.

4. Formed Elements of Blood - Overview

The formed elements are the cellular components suspended in plasma. They are red blood cells, white blood cells and platelets. Each has a specialised function: RBCs are concerned mainly with gas transport, WBCs with defense and immunity, and platelets with hemostasis and repair of damaged vessels.

Formed element	Normal value	Main function	Important point
RBC	Male: about 5 million/cu mm; female: about 4.5 million/cu mm	Transport of O ₂ and CO ₂ by hemoglobin.	Biconcave, non-nucleated, lifespan about 120 days.
WBC	4,000-11,000/cu mm	Defense, phagocytosis, immunity and inflammatory response.	Nucleated; granulocytes and agranulocytes.
Platelets	2,00,000-4,00,000/cu mm; average 2,50,000/cu mm	Hemostasis, clotting, clot retraction and vessel repair.	Cytoplasmic fragments from megakaryocytes; lifespan 8-11 days.

5. Red Blood Cells (Erythrocytes)

Red blood cells are non-nucleated formed elements of blood. Their red colour is due to hemoglobin. Compared with WBCs and platelets, RBCs are present in the largest number and are designed for the transport of respiratory gases. In humans, mature RBCs do not contain nucleus, mitochondria or Golgi apparatus, so their energy is produced mainly by glycolysis.

A normal RBC is biconcave and disc-shaped. The biconcave shape is highly useful because it provides a large surface area for diffusion, permits rapid movement of gases, gives flexibility during passage through narrow capillaries, and prevents excessive membrane tension during changes in volume. The average diameter is about 7.2 micrometre and the volume is about 85-90 cubic micrometre.

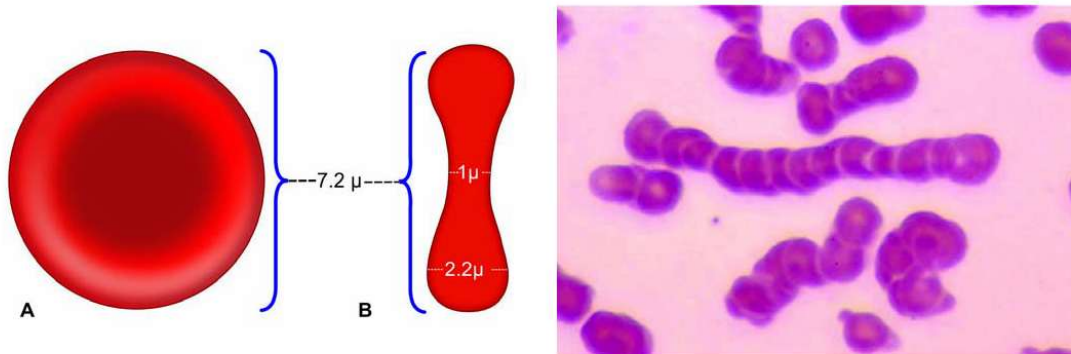


Figure: RBC dimensions and rouleaux formation. Source: Sembulingam.

Diagram tip: Rouleaux means RBCs pile up like coins. It is promoted by plasma proteins such as fibrinogen and globulins.

RBC feature	Exam value / fact
Shape	Biconcave disc.
Diameter	About 7.2 micrometre.
Thickness	Peripheral thickness about 2.2 micrometre; central thickness about 1 micrometre.
Surface area	About 120 square micrometre.
Volume	About 85-90 cubic micrometre.
Lifespan	About 120 days.
Site of destruction	Reticuloendothelial system, especially spleen.

Functions of RBCs are mainly related to hemoglobin. They transport oxygen as oxyhemoglobin and carbon dioxide partly as carbamino hemoglobin. Hemoglobin also acts as an important buffer and helps in acid-base balance. RBCs also help maintain the viscosity of blood and contribute to blood group antigen expression on the cell membrane.

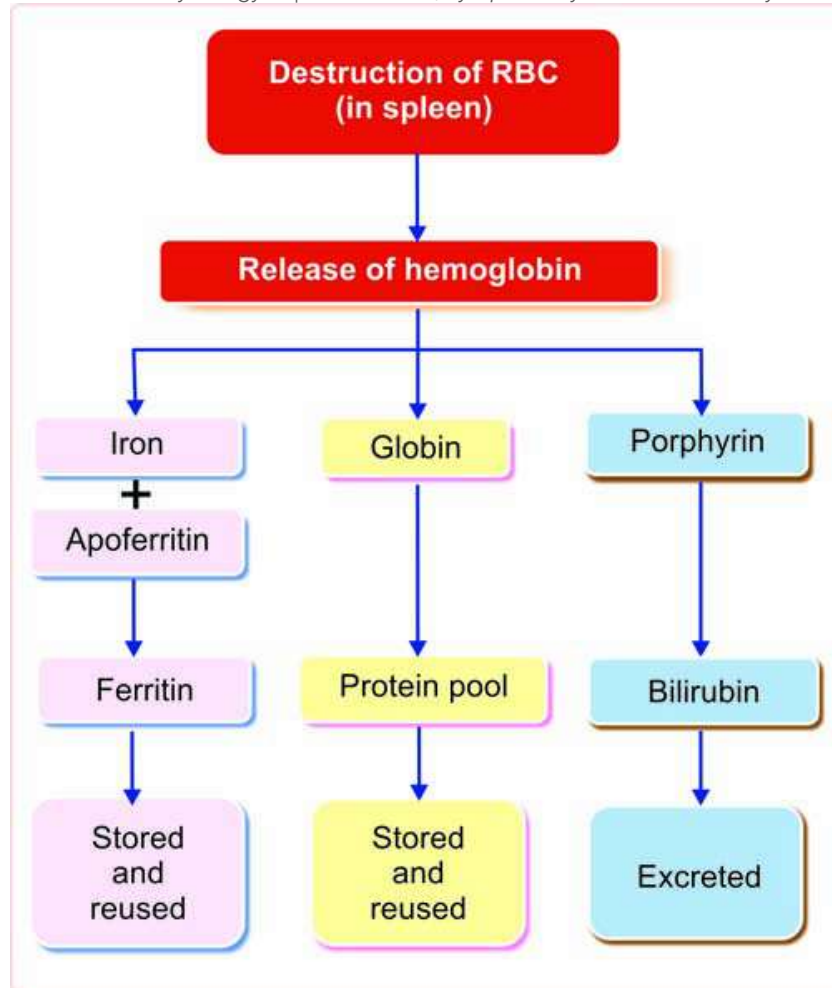


Figure: Fate of red blood cells after their lifespan. Source: Sembulingam.

Diagram tip: In answers on RBC destruction, mention: globin is reused, iron is stored as ferritin/hemosiderin and reused, and porphyrin becomes bile pigments.

6. Erythropoiesis - Formation of RBCs

Erythropoiesis is the process of formation of red blood cells. In adults, it occurs mainly in red bone marrow. The process begins from pluripotent hematopoietic stem cells and passes through committed erythroid progenitors, proerythroblast, normoblast stages, reticulocyte and finally mature erythrocyte. The nucleus is lost during maturation, and hemoglobin content progressively increases.

The most important regulator of erythropoiesis is erythropoietin, mainly secreted by the kidneys in response to hypoxia. When oxygen delivery to tissues decreases, erythropoietin secretion increases and stimulates red bone marrow to produce more RBCs. This is why chronic high altitude exposure causes physiological polycythemia.

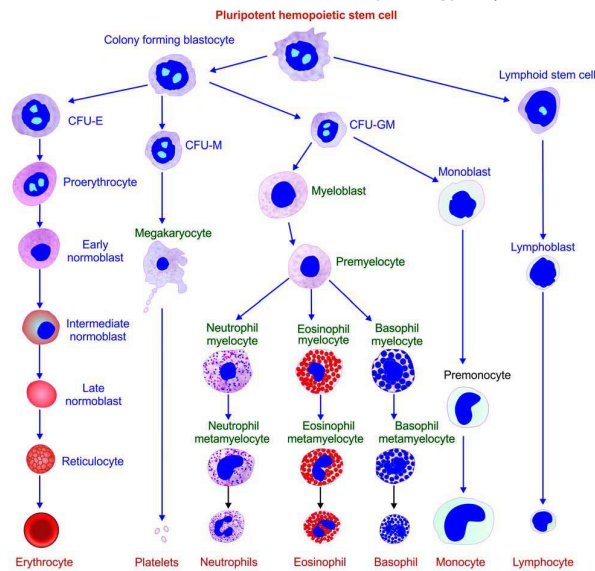


Figure: Stages of erythropoiesis. Source: Sembulingam.

Substances required for erythropoiesis

- Proteins and amino acids are needed for globin synthesis and cell growth.
- Iron is essential for heme formation and hemoglobin synthesis.
- Vitamin B12 and folic acid are required for DNA synthesis and nuclear maturation.
- Vitamin C helps iron absorption and folate metabolism.
- Copper and cobalt act as supportive factors in hemoglobin formation.
- Hormones such as thyroxine, androgens and growth hormone support marrow activity.

Applied point: Deficiency of vitamin B12 or folic acid causes defective DNA synthesis and produces megaloblastic anemia. Iron deficiency causes microcytic hypochromic anemia.

7. Hemoglobin - Structure, Functions and Synthesis

Hemoglobin is the iron-containing colouring pigment of RBCs. It is a chromoprotein and forms about 95% of the dry weight of RBC. Its molecular weight is about 68,000. The average hemoglobin content is 14-16 g/dL, about 15 g/dL in adult males and about 14.5 g/dL in adult females.

Structurally, hemoglobin is made of heme and globin. Heme contains iron in the ferrous state and a porphyrin ring. Globin is the protein part and contains four polypeptide chains. Adult hemoglobin has two alpha and two beta chains. Fetal hemoglobin has two alpha and two gamma chains, and it has higher affinity for oxygen than adult hemoglobin.

Function of hemoglobin	Explanation
Transport of oxygen	Oxygen combines reversibly with hemoglobin to form oxyhemoglobin. Iron remains in ferrous state.
Transport of carbon dioxide	CO ₂ combines with globin part to form carbaminohemoglobin.
Buffer action	Hemoglobin buffers hydrogen ions and helps maintain acid-base balance.
Colour of blood	Oxygenated Hb gives bright red colour; deoxygenated Hb gives darker venous colour.

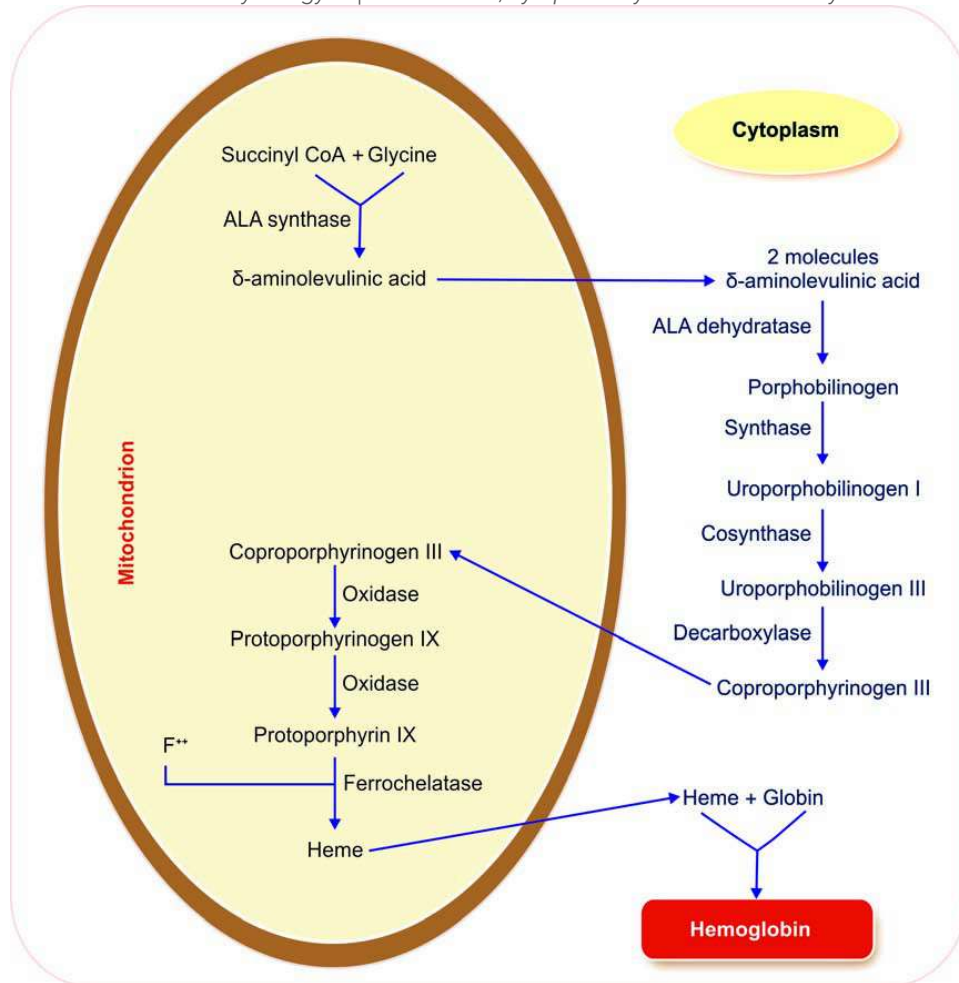


Figure: Synthesis of hemoglobin. Source: Sembulingam.

Diagram tip: Remember: heme is synthesized mainly in mitochondria; globin is synthesized by ribosomes; hemoglobin appears from intermediate normoblast stage.

Hemoglobin synthesis begins in the proerythroblast stage and continues up to the reticulocyte stage. Heme is formed from succinyl-CoA and glycine through porphyrin intermediates. Iron is inserted into protoporphyrin IX to form heme. Globin chains are synthesized in ribosomes and combine with heme to form hemoglobin.

After RBC destruction, hemoglobin is split into heme and globin. Globin is reused as amino acids. Heme is split into iron and porphyrin. Iron is stored as ferritin and hemosiderin and reused for hemoglobin synthesis. Porphyrin is converted to biliverdin and bilirubin, which are bile pigments.

Abnormal Hb derivative	Cause / importance
Carboxyhemoglobin	Formed when carbon monoxide combines with hemoglobin; reduces oxygen-carrying capacity.
Methemoglobin	Iron is oxidized from ferrous to ferric state; causes tissue hypoxia and cyanosis.
Sulfhemoglobin	Formed by combination with hydrogen sulfide; cannot be reconverted to hemoglobin.

8. Iron Metabolism - Exam Essentials

Iron is required for formation of hemoglobin, myoglobin, cytochromes, cytochrome oxidase, peroxidase and catalase. The total iron content of the body is about 4 g. Most of the body iron is present in hemoglobin, while a significant amount is stored in reticuloendothelial cells and liver.

Location / form	Approximate distribution
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Hemoglobin	65-68% of total iron.
Myoglobin in muscle	About 4%.
Intracellular oxidative heme compounds	About 1%.
Plasma transferrin	About 0.1%.
Storage in reticuloendothelial system	25-30%.

Dietary iron is present as heme iron and non-heme iron. Heme iron from fish, meat and chicken is absorbed more easily. Non-heme iron from vegetables, grains and cereals is absorbed less efficiently. Iron is absorbed mainly from the small intestine. In blood, iron is transported by transferrin and stored as ferritin or hemosiderin.

Clinical link: One gram of hemoglobin contains about 3.34 mg of iron. Loss of 100 mL blood causes loss of about 50 mg iron. This explains why chronic blood loss commonly causes iron deficiency anemia.

9. Anaemia

Anaemia is a blood disorder characterized by reduction in RBC count, hemoglobin content and/or packed cell volume. It may occur due to decreased production of RBCs, increased destruction of RBCs, or excessive blood loss. The final result is reduced oxygen-carrying capacity of blood, leading to tissue hypoxia.

Morphological classification is based on size and colour of RBCs. Size is assessed by mean corpuscular volume and colour by mean corpuscular hemoglobin concentration. Etiological classification is based on cause, such as hemorrhage, hemolysis, nutritional deficiency, bone marrow failure and chronic disease.

Morphological type	RBC appearance	Common example
Normocytic normochromic	Normal size and normal colour, but reduced number.	Acute blood loss, aplastic anemia, anemia of chronic disease.
Macrocytic normochromic	Large RBCs with normal colour.	Vitamin B12 deficiency, folate deficiency.
Macrocytic hypochromic	Large pale RBCs.	Some nutritional deficiency states.
Microcytic hypochromic	Small pale RBCs.	Iron deficiency anemia, thalassemia.

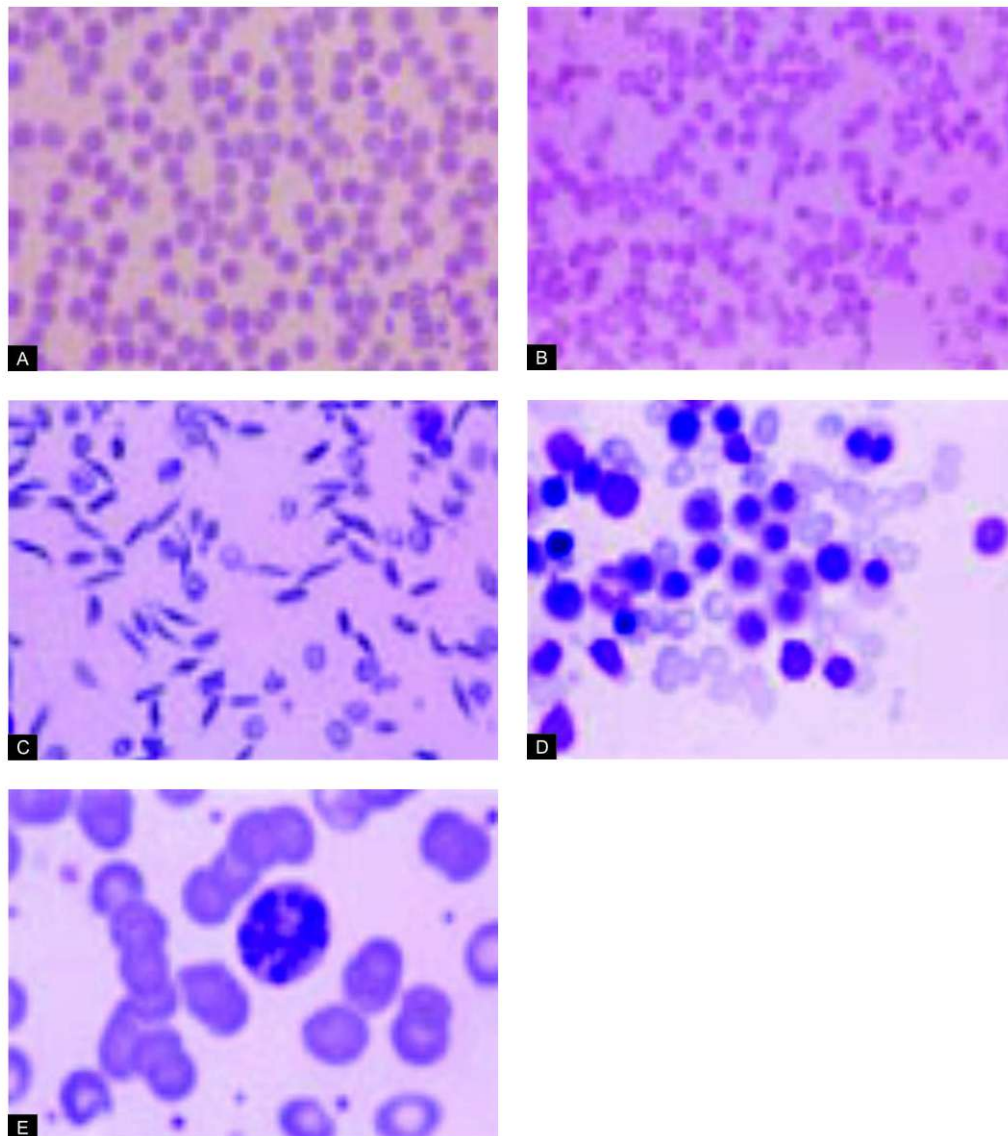


Figure: Normal RBC and common RBC appearances in anaemia. Source: Sembulingam.

Diagram tip: In anemia questions, add RBC morphology. It instantly shows whether you know the classification.

Etiological type	Cause	Typical morphology
Hemorrhagic anemia	Acute or chronic blood loss.	Acute: normocytic normochromic; chronic: microcytic hypochromic.
Hemolytic anemia	Excessive RBC destruction due to intrinsic or extrinsic causes.	Often normocytic normochromic; sickle shape in sickle cell anemia.
Nutritional deficiency anemia	Deficiency of iron, protein, vitamin B12 or folic acid.	Iron deficiency: microcytic hypochromic; B12/folate: macrocytic/megaloblastic.
Aplastic anemia	Bone marrow failure.	Normocytic normochromic.
Anemia of chronic disease	Chronic infection, inflammatory disease, renal failure or malignancy.	Usually normocytic normochromic.

Common signs and symptoms include pallor, fatigue, weakness, breathlessness on exertion, palpitations, dizziness, headache, tachycardia and reduced exercise tolerance. Severe anemia may

cause cardiac strain because the heart attempts to compensate for reduced oxygen-carrying capacity by increasing cardiac output.

Blood index	Meaning	Normal value / importance
MCV	Mean corpuscular volume: average volume of one RBC.	About 78-90 cubic micrometre; identifies microcytic or macrocytic anemia.
MCH	Mean corpuscular hemoglobin: amount of Hb in one RBC.	About 27-32 pg.
MCHC	Mean corpuscular hemoglobin concentration: Hb concentration in RBC.	About 30-38%; most important for hypochromia.
Colour index	Ratio of Hb percentage to RBC percentage.	Reduced in iron deficiency anemia; increased in macrocytic anemia.

10. White Blood Cells (Leukocytes)

White blood cells are colourless, nucleated formed elements of blood. They are fewer in number than RBCs but are essential for defense. WBCs protect the body from invading organisms by phagocytosis, immune reactions, allergic responses, inflammation and removal of damaged tissue.

WBCs are classified into granulocytes and agranulocytes. Granulocytes have cytoplasmic granules and include neutrophils, eosinophils and basophils. Agranulocytes lack obvious granules and include monocytes and lymphocytes. The normal total WBC count is 4,000-11,000/cu mm of blood.

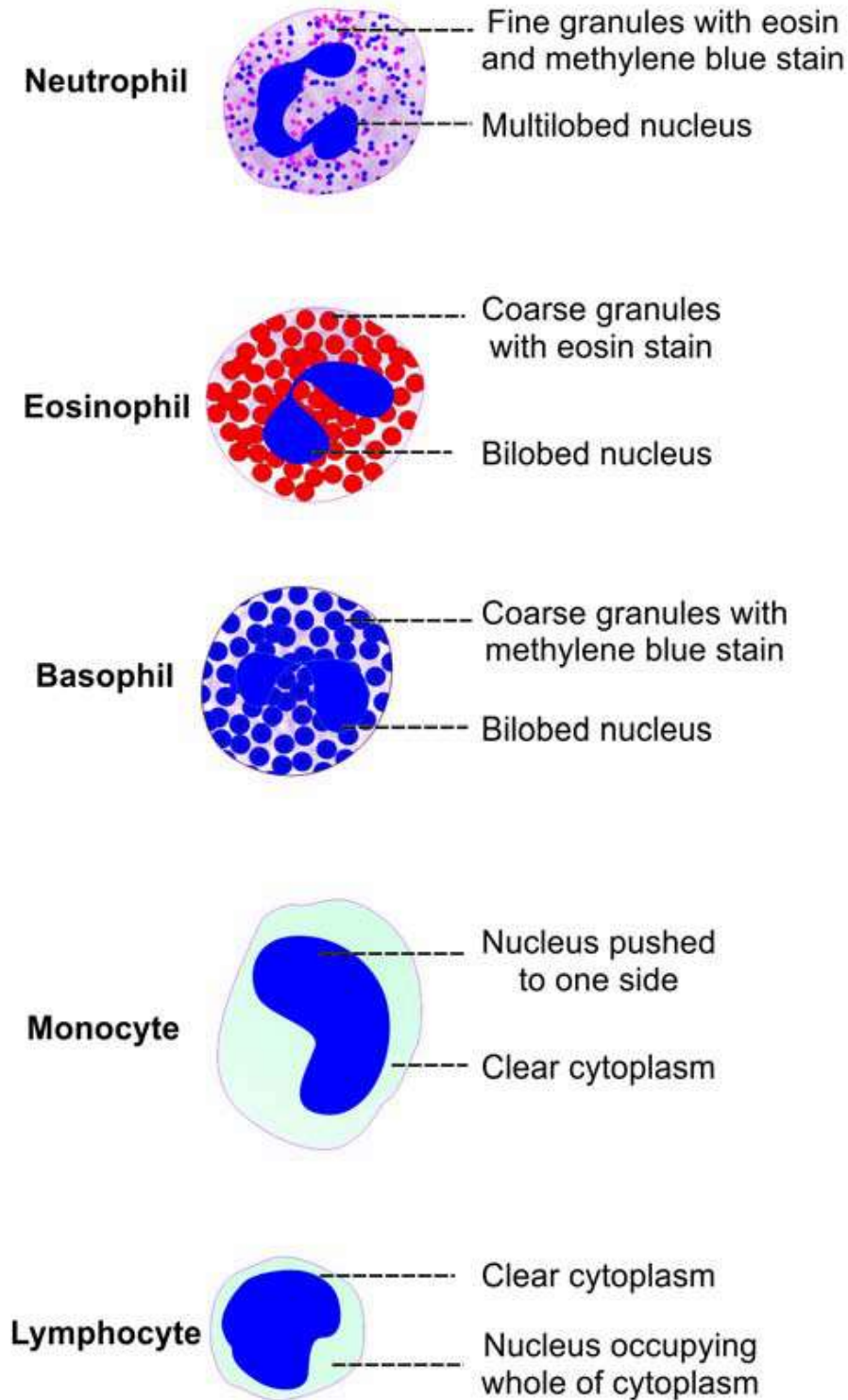


Figure: Different white blood cells with identifying features. Source: Sembulingam.

Diagram tip: For diagrams, label nucleus shape and granules. Neutrophil = multilobed, eosinophil = bilobed red granules, basophil = blue granules, monocyte = kidney-shaped nucleus, lymphocyte = large nucleus.

WBC	Normal %	Main function
Neutrophils	50-70%	First line cellular defense; phagocytosis of bacteria; acute inflammation.
Eosinophils	2-4%	Defense against parasites; control of allergic responses; detoxification of foreign proteins.
Basophils	0-1%	Release heparin, histamine and other mediators; allergic and

		inflammatory reactions.
Monocytes	2-6%	Become macrophages in tissues; phagocytosis and antigen presentation.
Lymphocytes	20-30%	B cells produce antibodies; T cells provide cellular immunity; memory cells give long-term immunity.

11. Properties and Functions of WBCs

The main functional properties of WBCs are diapedesis, amoeboid movement, chemotaxis and phagocytosis. Diapedesis means passage of WBCs through intact capillary walls. Amoeboid movement helps them move through tissues. Chemotaxis is movement towards chemicals released at injured or infected areas. Phagocytosis is engulfing and destroying microorganisms or foreign particles.

Property	Meaning	Importance
Diapedesis	WBCs squeeze through capillary walls.	Allows WBCs to leave blood and enter infected tissue.
Amoeboid movement	Movement by changing shape.	Helps WBC migration through tissues.
Chemotaxis	Directed movement towards chemical attractants.	Brings WBCs to site of injury/infection.
Phagocytosis	Engulfing and digestion of foreign particles.	Main action of neutrophils and macrophages.

Neutrophils are particularly important in acute bacterial infections. Monocytes migrate into tissues and become macrophages, which are more powerful phagocytes and also present antigens to lymphocytes. Lymphocytes are the central cells of acquired immunity. Eosinophils are important in parasitic infections and allergic conditions, whereas basophils participate in allergic reactions by releasing histamine and heparin-like substances.

MCQ tip: The sequence in inflammation/phagocytosis-type questions is commonly: chemotaxis -> diapedesis -> opsonization/recognition -> phagocytosis -> digestion.

12. Platelets

Platelets, or thrombocytes, are small colourless, non-nucleated fragments of cytoplasm derived from megakaryocytes in bone marrow. Their normal count is about 2,50,000/cu mm, with a usual range of 2,00,000-4,00,000/cu mm. They are inactive in normal circulation but become activated when they contact damaged endothelium and exposed collagen.

Feature	Point
Size	Diameter about 2.5 micrometre; volume about 7.5 cubic micrometre.
Shape	Inactive platelets are usually disc-shaped; activated platelets form filopodia.
Lifespan	About 8-11 days, average around 10 days.
Fate	Destroyed mainly by tissue macrophage system in spleen.
Important proteins	Actin, myosin, thrombosthenin, von Willebrand factor, platelet-derived growth factor.

Platelets have three important properties, often remembered as the three As: adhesiveness, aggregation and agglutination. Adhesiveness is sticking to collagen at the site of injury. Aggregation is grouping of activated platelets. Agglutination is clumping together of platelets. These reactions produce a temporary platelet plug and begin the process of hemostasis.

Function of platelets	Explanation
Blood clotting	Provide platelet phospholipid and help form intrinsic prothrombin activator.
Clot retraction	Actin, myosin and thrombosthenin contract the clot and squeeze out serum.
Prevention of blood loss	Secrete serotonin, adhere to damaged areas and form temporary platelet plug.

Vessel repair	Platelet-derived growth factor supports endothelial repair and wound healing.
Defense	Agglutinate around foreign bodies and help their removal.

13. Hemostasis

Hemostasis means arrest or stoppage of bleeding. It begins immediately after vascular injury and prevents excessive blood loss. The process has three major stages: vasoconstriction, platelet plug formation and coagulation. These stages overlap with one another and are reinforced by local chemical mediators released from damaged tissue, endothelium and platelets.

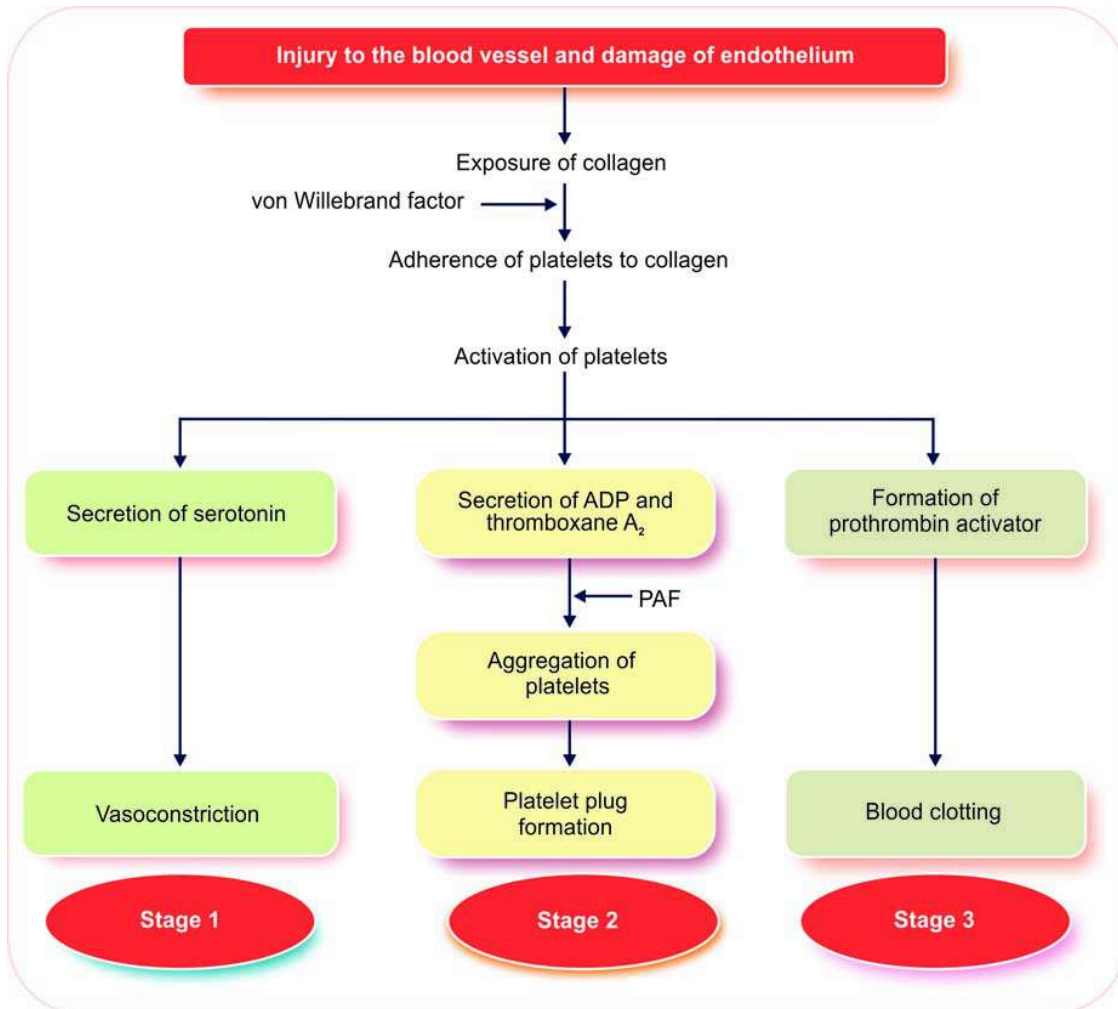


Figure: Stages of hemostasis. Source: Sembulingam.

Diagram tip: For short notes on hemostasis, always write the three stages in order: vasoconstriction -> platelet plug -> coagulation.

In vasoconstriction, the injured vessel constricts reflexly and due to local substances such as serotonin and thromboxane A₂, reducing blood flow. In platelet plug formation, platelets adhere to exposed collagen through von Willebrand factor, become activated, release ADP and thromboxane A₂, and aggregate to form a temporary plug. In coagulation, fibrin threads form and stabilize the platelet plug into a firm clot.

Stage	Key event	Result
1. Vasoconstriction	Constriction of injured vessel due to myogenic spasm and local mediators.	Immediate reduction in blood loss.
2. Platelet plug	Platelet adhesion, activation, release reaction and aggregation.	Temporary plug seals small openings.
3. Coagulation	Activation of clotting cascade and fibrin formation.	Stable clot forms.

14. Coagulation of Blood

Coagulation, or clotting, is the process by which blood loses its fluidity and becomes a jelly-like mass after it is shed or collected in a container. It is a cascade of enzyme reactions. Most clotting factors circulate as inactive proenzymes and become activated step by step until thrombin is formed. Thrombin then converts fibrinogen into fibrin.

Factor	Name
I	Fibrinogen
II	Prothrombin
III	Tissue thromboplastin / tissue factor
IV	Calcium
V	Labile factor / proaccelerin
VI	Presence not proved
VII	Stable factor
VIII	Antihemophilic factor
IX	Christmas factor
X	Stuart-Prower factor
XI	Plasma thromboplastin antecedent
XII	Hageman factor
XIII	Fibrin-stabilizing factor

Important coagulation factors are easier to remember by function. Factor I is the soluble fibrinogen that becomes fibrin. Factor II is prothrombin, the precursor of thrombin. Factor III is tissue factor from damaged tissues. Factor IV is calcium, which is required at many steps. Factor XIII stabilizes the final fibrin meshwork.

The liver synthesizes most coagulation factors. Vitamin K is required for normal synthesis of factors II, VII, IX and X, so vitamin K deficiency, obstructive jaundice, severe liver disease and warfarin therapy may prolong clotting. Calcium is essential for clotting, but in a living body blood does not clot inside vessels because of intact endothelium, continuous flow and natural anticoagulants.

Test / point	What it mainly assesses
Bleeding time	Platelet plug formation and capillary function; prolonged in platelet disorders.
Clotting time	Overall ability of blood to clot; prolonged in coagulation factor deficiency.
Prothrombin time (PT)	Extrinsic and common pathways; useful in warfarin monitoring.
Activated partial thromboplastin time (aPTT)	Intrinsic and common pathways; prolonged in hemophilia and heparin effect.
Clot retraction	Platelet contractile function and fibrin meshwork stability.

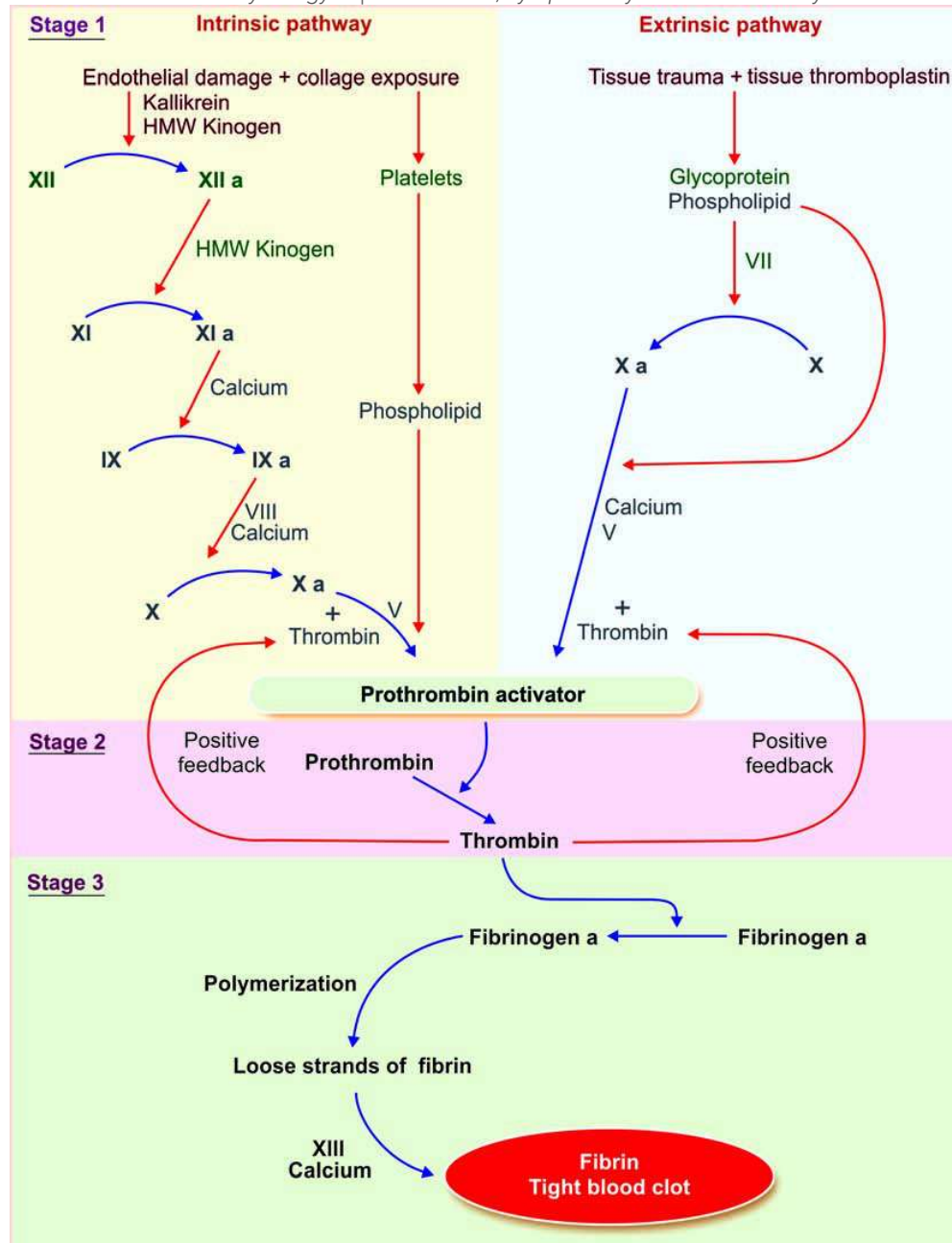


Figure: Stages of blood coagulation. Source: Sembulingam.

Diagram tip: Common exam answer: intrinsic and extrinsic pathways both activate factor X. Activated factor X with factor V, phospholipid and calcium forms prothrombin activator.

The three major stages of clotting are: formation of prothrombin activator, conversion of prothrombin into thrombin, and conversion of fibrinogen into fibrin. Prothrombin activator is formed by intrinsic and extrinsic pathways. The intrinsic pathway starts within blood after contact of factor XII with collagen. The extrinsic pathway starts outside blood when injured tissue releases tissue thromboplastin.

Pathway	Initiated by	Important sequence	Exam point
Intrinsic	Blood components and platelets; exposure of collagen after endothelial injury.	XII → XI → IX; IX with VIII and Ca ²⁺ activates X.	Slower pathway; all factors are within blood.
Extrinsic	Tissue thromboplastin released from injured tissues.	Tissue factor with VII activates X.	Faster pathway; begins outside blood.
Common pathway	Activated factor X.	Xa + V + phospholipid + Ca ²⁺ forms prothrombin activator; prothrombin -	Factor XIII stabilizes fibrin

		> thrombin; fibrinogen -> fibrin.	clot.
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After a clot forms, clot retraction begins. Platelet contractile proteins such as actin, myosin and thrombosthenin contract the clot, and serum oozes out. Later, fibrinolysis removes the clot from the vessel lumen. Plasminogen is converted into plasmin by tissue plasminogen activator, urokinase and other activators. Plasmin digests fibrin.

Intravascular clotting is normally prevented by continuous blood flow, smooth endothelial lining, inactive clotting factors and natural anticoagulants such as heparin, antithrombin III, thrombomodulin, protein C and protein S. Clinically used anticoagulants include heparin, warfarin/coumarin derivatives, oxalates and citrates depending on the purpose.

Bleeding disorders: Hemophilia A is due to deficiency of factor VIII; hemophilia B is due to deficiency of factor IX. Vitamin K is required for synthesis of factors II, VII, IX and X.

15. Blood Groups - ABO System

Blood groups are determined by antigens present on the surface of RBCs. In the ABO system, the important antigens are A and B, also called agglutinogens. Plasma contains naturally occurring antibodies called agglutinins: anti-A and anti-B. A person cannot normally have an antigen and its corresponding antibody together because this would cause agglutination.

Blood group	Antigen on RBC	Antibody in plasma	Can donate RBCs to	Can receive RBCs from
A	A	Anti-B	A, AB	A, O
B	B	Anti-A	B, AB	B, O
AB	A and B	None	AB	A, B, AB, O
O	None	Anti-A and Anti-B	A, B, AB, O	O

Landsteiner law states that if a particular agglutinin is present in the RBCs, the corresponding agglutinin must be absent from serum. If the agglutinin is absent, the corresponding agglutinin is usually present. The second part of this law is not applicable to Rh factor because anti-D antibodies are not naturally present; they develop after exposure.

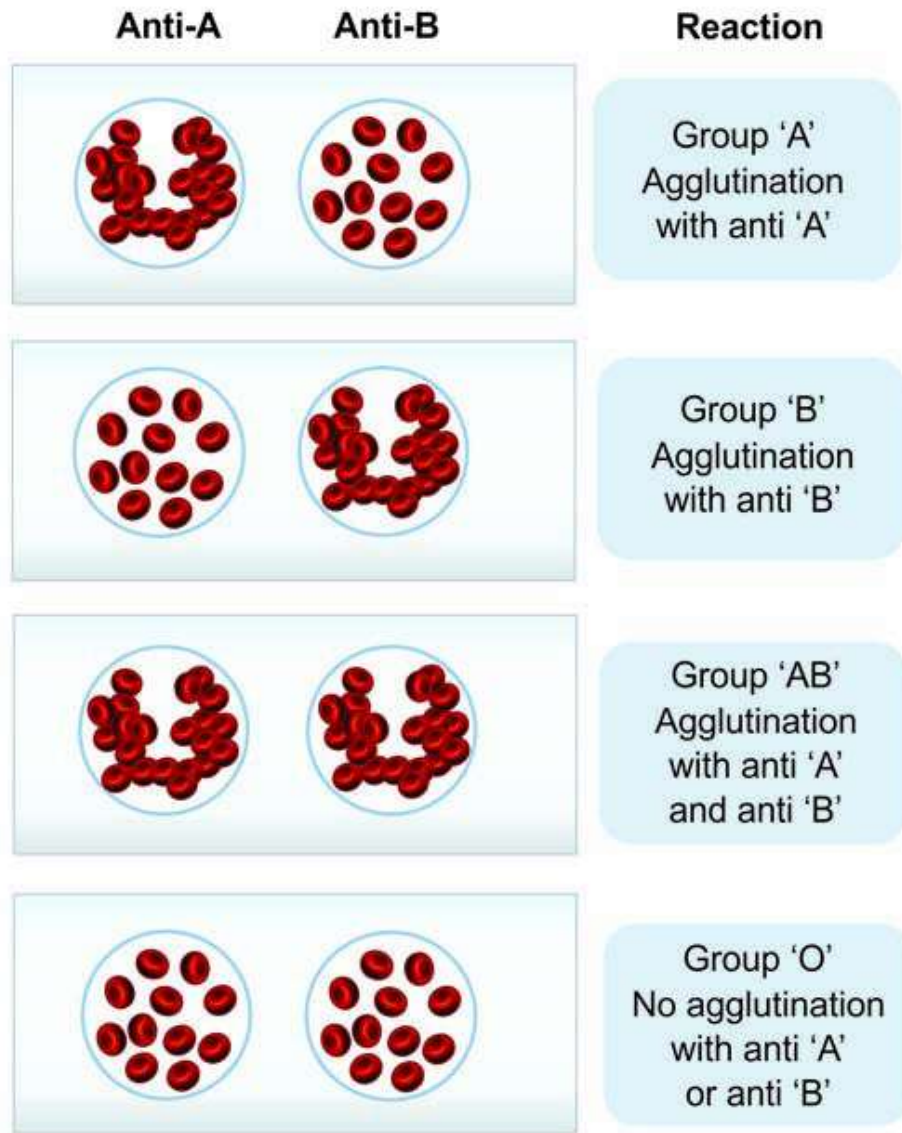


Figure: Determination of ABO blood group by agglutination. Source: Sembulingam.

Diagram tip: Agglutination with anti-A means A antigen is present. Agglutination with anti-B means B antigen is present. No agglutination with both means group O.

16. Rh Factor and Blood Transfusion

The Rh system is another important blood group system. The most important Rh antigen is D antigen. If D antigen is present, the person is Rh positive. If D antigen is absent, the person is Rh negative. Unlike ABO antibodies, anti-D antibodies are not naturally present in Rh-negative individuals. They develop only after exposure to Rh-positive blood or Rh-positive fetal RBCs.

Before blood transfusion, donor and recipient blood must be matched and cross-matched. Major cross-match tests donor RBCs against recipient serum. Minor cross-match tests donor serum against recipient RBCs. The purpose is to prevent agglutination and hemolysis due to incompatible antibodies.

Transfusion reaction	Mechanism	Important effects
ABO incompatibility	Recipient antibodies attack donor RBC antigens.	Agglutination, hemolysis, fever, renal damage, shock and possible death.
Rh incompatibility	Anti-D antibodies form after exposure and attack Rh-positive RBCs on later exposure.	Hemolytic transfusion reaction or hemolytic disease of fetus/newborn.

Hemolytic disease of newborn	Rh-negative mother forms anti-D against Rh-positive fetus; antibodies cross placenta in later pregnancy.	Fetal RBC destruction, anemia, jaundice, erythroblastosis fetalis.
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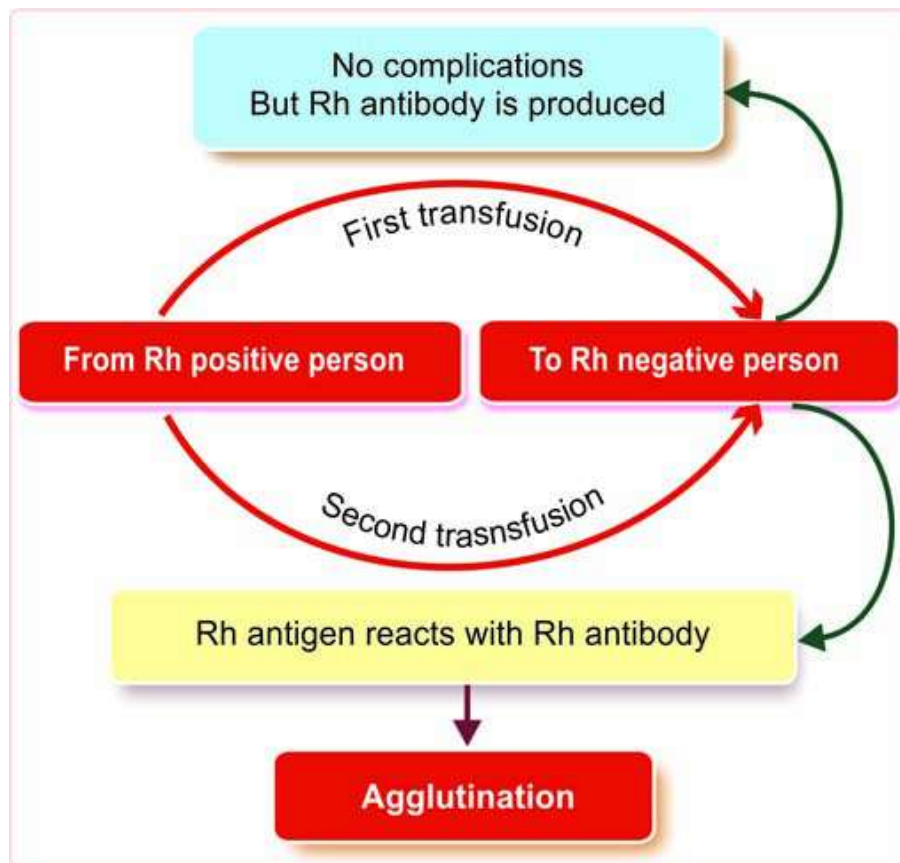


Figure: Rh incompatibility. Source: Sembulingam.

Diagram tip: Rh-negative mother + Rh-positive fetus is the classic setup. Problem usually appears in later pregnancy after sensitization.

17. Lymphatic System and Lymph

The lymphatic system is a closed system of lymph channels through which lymph flows from tissue spaces back to blood. It is a one-way drainage system. Lymphatic capillaries begin as blind-ended terminals in tissue spaces. They are more porous than blood capillaries and allow excess tissue fluid, proteins and absorbed lipids to enter.

Lymph vessels unite to form larger vessels and finally drain into the venous system. The right lymphatic duct drains the right upper part of the body and opens into the right subclavian vein. The thoracic duct drains the rest of the body and opens into the left subclavian vein. This drainage is important for maintaining interstitial fluid volume and returning plasma proteins to blood.

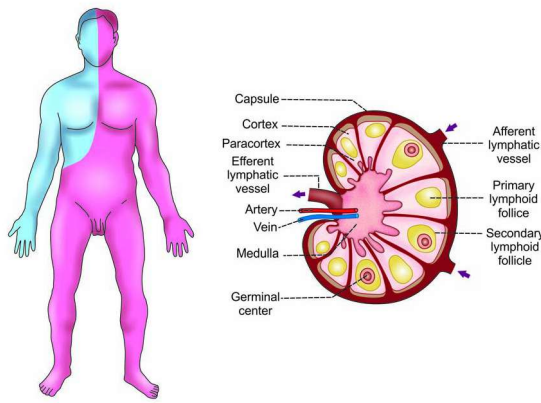


Figure: Lymph drainage and lymph node. Source: Sembulingam.

Important lymphatic facts

- Lymph flows from tissue spaces towards blood only.
- Valves prevent backflow of lymph.
- Lymph nodes are filters placed along lymph vessels.
- Cortex contains B lymphocytes and follicles; paracortex contains T lymphocytes.
- Medulla contains lymphocytes, macrophages and blood vessels.
- Swollen painful lymph nodes commonly indicate infection in the drainage area.

18. Composition and Functions of Lymph

Lymph is the fluid present in lymphatic vessels. It is derived from tissue fluid and therefore resembles plasma, but the protein concentration is usually lower. Lymph contains water, electrolytes, lymphocytes, a small amount of proteins, absorbed fats from intestine and waste products from tissues. Intestinal lymph after fat absorption is called chyle because it becomes milky due to chylomicrons.

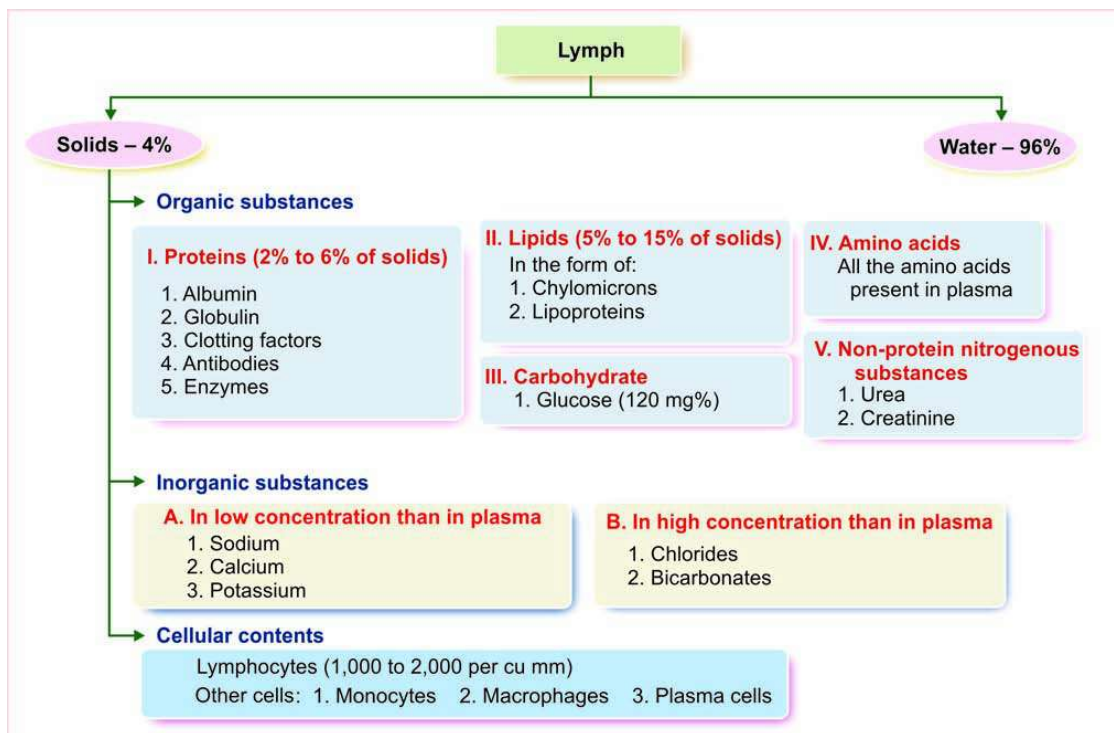


Figure: Composition of lymph. Source: Sembulingam.

Diagram tip: For composition, write: lymph = water + electrolytes + proteins + lymphocytes + absorbed fats + waste products. Intestinal lymph is rich in fat.

Function of lymph	Explanation
Return of proteins	Returns plasma proteins from interstitial spaces back to blood.
Fluid balance	Drains excess tissue fluid and prevents edema.
Fat absorption	Lacteals absorb fats from small intestine and carry them as chyle.
Defense	Lymph carries antigens to lymph nodes where immune responses are initiated.
Transport of lymphocytes	Allows circulation of lymphocytes between lymphoid tissue and blood.

Lymph nodes act as biological filters. As lymph passes through them, bacteria, foreign particles and toxic substances are trapped and destroyed by macrophages. Lymph nodes are therefore called defense barriers. During infection, local lymph nodes become enlarged because lymphocytes and macrophages proliferate and become more active.

19. Lymphatic Tissue and Lymphoid Organs

Lymphatic tissue is a specialized connective tissue rich in lymphocytes. It provides the structural site where immune cells are produced, matured, stored and activated. Lymphoid organs are classified into primary and secondary organs. Primary lymphoid organs are involved in maturation of lymphocytes, while secondary lymphoid organs are sites where lymphocytes meet antigens and mount immune responses.

Type	Organs / tissues	Main role
Primary lymphoid organs	Bone marrow and thymus.	Production and maturation of lymphocytes. B cells mature in bone marrow; T cells mature in thymus.
Secondary lymphoid organs	Lymph nodes, spleen, tonsils, Peyer patches, appendix and mucosa-associated lymphoid tissue.	Trap antigens and provide sites for immune response.
Diffuse lymphatic tissue	Scattered lymphocytes in mucosa of respiratory, digestive and urinary tracts.	Local defense at body surfaces.

The spleen is an important lymphoid organ related to blood rather than lymph. It filters blood, removes old RBCs and platelets, stores blood in some animals, and participates in immune responses. Lymph nodes filter lymph, while spleen filters blood. This difference is frequently tested in exams.

Difference to remember: Lymph node filters lymph. Spleen filters blood. Thymus trains T lymphocytes. Bone marrow forms blood cells and processes B lymphocytes after birth.

20. Immunity - Innate and Acquired

Immunity is the capacity of the body to resist pathogenic agents such as bacteria, viruses, toxins and other foreign substances. It is of two major types: innate immunity and acquired immunity. Innate immunity is present from birth and provides non-specific defense. Acquired immunity develops after exposure to specific antigens and has specificity and memory.

Feature	Innate immunity	Acquired immunity
Specificity	Non-specific.	Specific to antigen.
Onset	Immediate.	Slower during first exposure.
Memory	Absent or minimal.	Present; second response is faster and stronger.
Main components	Skin, mucosa, phagocytes, complement, NK cells, interferons.	T lymphocytes, B lymphocytes, antibodies and memory cells.
Example	Neutrophil phagocytosis of bacteria.	Antibody production after vaccination.

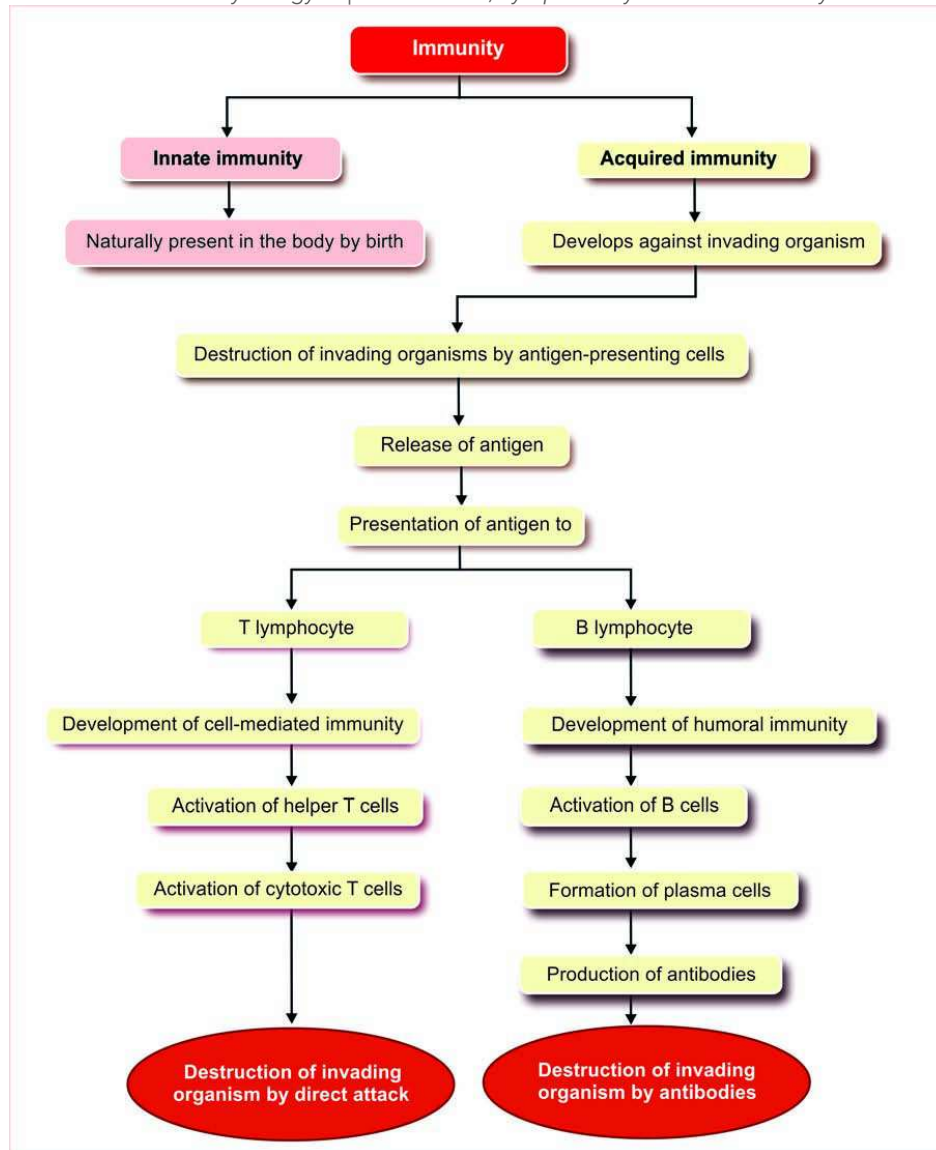


Figure: Schematic diagram showing development of immunity. Source: Sembulingam.

Diagram tip: Acquired immunity divides into cellular immunity by T cells and humoral immunity by B cells/antibodies.

Cell-mediated immunity is mediated mainly by T lymphocytes and does not depend on circulating antibodies. It is important against viruses, fungi, some intracellular bacteria, transplanted tissues and tumour cells. Humoral immunity is mediated by B lymphocytes, plasma cells and antibodies. It is especially important against extracellular bacteria, toxins and many soluble antigens.

21. Role of Thymus in Immunity

The thymus is a primary lymphoid organ essential for processing and maturation of T lymphocytes. T lymphocytes are processed mostly during the period just before birth and in the first few months after birth. The thymus secretes thymosin, which accelerates proliferation and activation of lymphocytes in the thymus and increases lymphocyte activity in lymphoid tissues.

During processing in the thymus, immature lymphocytes become functionally competent T cells. They differentiate into helper T cells, cytotoxic T cells, suppressor/regulatory T cells and memory T cells. Helper T cells are also called CD4 cells and coordinate immune reactions. Cytotoxic T cells are also called CD8 cells and destroy virus-infected cells, tumour cells and foreign graft cells.

T cell type	Main function
Helper T cell / CD4	Activates B cells, cytotoxic T cells and macrophages; secretes cytokines;

	coordinates immune response.
Cytotoxic T cell / CD8	Kills virus-infected cells, tumour cells and graft cells.
Suppressor / regulatory T cell	Controls and limits immune response; helps prevent autoimmunity.
Memory T cell	Persists after first exposure and gives faster response on re-exposure.

After transformation, T lymphocytes leave the thymus and are stored in lymphoid tissues such as lymph nodes, spleen, bone marrow and gastrointestinal tract. If the thymus is absent or defective, T cell development becomes defective and cellular immunity is reduced. DiGeorge syndrome, caused by absence or poor development of thymus, is a classic example of congenital immune deficiency.

22. B Lymphocytes, Antibodies and Humoral Immunity

B lymphocytes are processed in liver during fetal life and in bone marrow after birth. After processing, they become plasma cells and memory B cells. Plasma cells secrete antibodies, also called immunoglobulins. Memory cells remain in the body and produce rapid antibody response when the same antigen enters again.

In humoral immunity, the antigen binds to specific receptors on B lymphocytes. With the help of helper T cells and cytokines, selected B cells multiply and differentiate into plasma cells. Plasma cells secrete antibodies into blood and lymph. These antibodies combine with the antigen and help its destruction directly or through complement activation.

Response	Primary immune response	Secondary immune response
Occurs when	First exposure to antigen.	Second or later exposure to same antigen.
Speed	Slow onset because new clones must be activated.	Rapid due to memory cells.
Antibody level	Lower antibody titre.	Higher antibody titre.
Main antibody	IgM appears first, followed by IgG.	IgG response is usually stronger and faster.
Importance	Basis of initial infection/vaccination response.	Basis of booster dose and long-term protection.

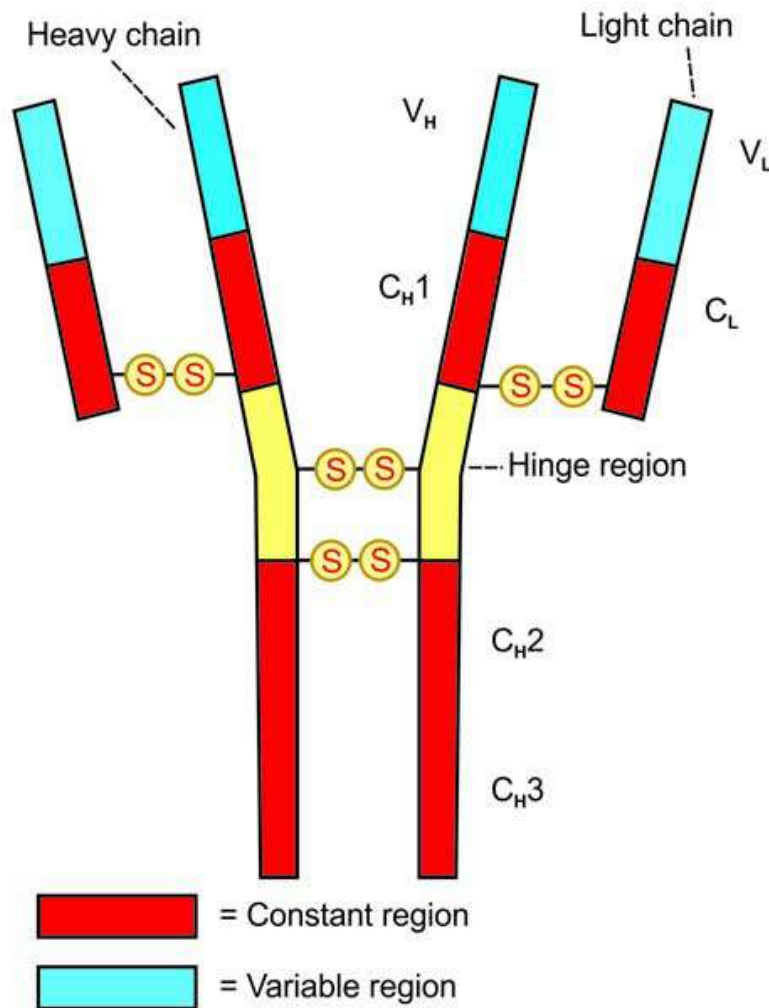


Figure: Structure of antibody molecule. Source: Sembulingam.

Diagram tip: For antibody diagrams, label heavy chains, light chains, variable region and constant region.

Immunoglobulin	High-yield function
IgG	Most abundant; crosses placenta; gives passive immunity to fetus.
IgA	Present in secretions such as saliva, tears, mucus and breast milk; mucosal protection.
IgM	First antibody produced in primary immune response; strong agglutination.
IgE	Allergy and parasitic infections; binds to mast cells and basophils.
IgD	Mainly present on surface of B lymphocytes; receptor function.

Antibodies protect by neutralization of toxins and viruses, agglutination of particles, precipitation of soluble antigens, opsonization to help phagocytosis, and activation of complement. Complement activation produces lysis of microbes, chemotaxis of leukocytes, opsonization, increased inflammation and enhanced clearance of immune complexes.

Memory aid: T cells = Thymus + cellular immunity. B cells = Bone marrow + antibodies. Plasma cells = antibody factories. Memory cells = faster second response.

23. Active and Passive Immunity

Acquired immunity may be active or passive. Active immunity develops when the person's own immune system responds to an antigen. Passive immunity is produced by transfer of ready-made antibodies from another source. Active immunity is slower but long-lasting; passive immunity is immediate but short-lived.

Type	How it develops	Examples	Duration
Active natural	After natural infection.	Immunity after recovery from measles or chickenpox.	Usually long-lasting.
Active artificial	After vaccination or toxoid administration.	BCG, hepatitis B vaccine, tetanus toxoid.	Long-lasting; may need boosters.
Passive natural	Maternal antibodies transferred to fetus or infant.	IgG through placenta; IgA through breast milk.	Short-lived.
Passive artificial	Injection of prepared antibodies.	Antiserum, immunoglobulin therapy.	Immediate but temporary.

Immunization is the procedure by which the body is prepared to fight against a specific disease. Vaccines contain killed organisms, live attenuated organisms, antigenic components or toxoids. Toxoids are toxins whose toxicity is destroyed but antigenicity is retained, such as tetanus and diphtheria toxoids.

24. Rapid Revision Tables

Topic	Must-write points
Composition of blood	Plasma 55%, formed elements 45%; formed elements = RBC, WBC, platelets; buffy coat = WBC + platelets.
Functions of blood	Nutritive, respiratory, excretory, hormone transport, water balance, acid-base balance, temperature regulation, storage, defense.
RBC	Biconcave, non-nucleated, 7.2 micrometre diameter, lifespan 120 days, Hb transport of gases.
Hemoglobin	Heme + globin; iron in ferrous state; transports O ₂ and CO ₂ ; buffer; adult Hb = $\alpha_2\beta_2$.
Anaemia	Reduced RBC count/Hb/PCV; classify by morphology and etiology; iron deficiency = microcytic hypochromic.
WBC	Neutrophils 50-70%, lymphocytes 20-30%, eosinophils 2-4%, monocytes 2-6%, basophils 0-1%.
Platelets	2,00,000-4,00,000/cu mm; adhesion, aggregation, agglutination; clotting and clot retraction.
Coagulation	Intrinsic and extrinsic pathways form prothrombin activator; thrombin converts fibrinogen to fibrin.
ABO blood group	A antigen has anti-B; B antigen has anti-A; AB has no antibody; O has both antibodies.
Lymph	Returns proteins, drains excess fluid, absorbs fats, supports immunity.
Thymus	Processes T lymphocytes; thymosin supports lymphocyte proliferation and activation.

25. Probable Exam Questions from Unit II

Long answer / essay	Short notes / 5-mark topics
Describe the composition and functions of blood.	Plasma vs serum; plasma proteins; buffy coat.
Describe RBC morphology, functions and fate.	Rouleaux formation; PCV; blood indices.
Describe hemoglobin structure, synthesis and functions.	Carboxyhemoglobin; fetal Hb; iron metabolism.
Classify anemia and describe important types.	Iron deficiency anemia; megaloblastic anemia; sickle cell anemia.
Classify WBCs and write their functions.	Properties of WBC; differential leukocyte count.
Describe hemostasis and coagulation of blood.	Intrinsic pathway; extrinsic pathway; clotting factors; fibrinolysis.
Explain ABO and Rh blood group systems.	Landsteiner law; cross matching; Rh incompatibility.
Describe lymphatic system and lymph nodes.	Composition of lymph; functions of lymph; lymph node structure.
Describe acquired immunity with role of thymus.	T cells; B cells; antibody structure; active/passive immunity.

26. One-page Memory Sheet

Blood values: Blood volume about 5 L, blood pH 7.4, plasma 55%, formed elements 45%, RBC count about 5 million/cu mm in males and 4.5 million/cu mm in females, WBC count 4,000-11,000/cu mm, platelet count 2,00,000-4,00,000/cu mm, Hb about 15 g/dL in adult male and 14.5 g/dL in adult female.

Clotting pathway memory: intrinsic begins with factor XII after contact with collagen; extrinsic begins with tissue factor III. Both activate factor X. Activated X with factor V, phospholipid and calcium forms prothrombin activator. Prothrombin becomes thrombin, and fibrinogen becomes fibrin. Factor XIII stabilizes fibrin.

Immunity memory: innate is non-specific and immediate; acquired is specific and has memory. T cells mature in thymus and produce cellular immunity. B cells mature in bone marrow and produce humoral immunity through antibodies. Lymph nodes filter lymph, spleen filters blood and thymus trains T cells.

Final writing pattern: Definition + normal value + classification/table + mechanism/diagram + applied point = full-scoring physiology answer.

Viva trigger	Best one-line answer
Why is blood called a connective tissue?	It has cells suspended in an extracellular matrix called plasma and connects body systems by transport.
Why does anemia cause breathlessness?	Reduced hemoglobin lowers oxygen delivery, so ventilation and cardiac output increase during activity.
Why is group O called universal donor for RBCs?	Its RBCs lack A and B antigens, so recipient anti-A/anti-B antibodies do not agglutinate them.
Why is AB called universal recipient for RBCs?	AB plasma lacks anti-A and anti-B antibodies, so donor RBCs are less likely to agglutinate.
What is the key role of thymus?	It processes T lymphocytes and supports cell-mediated immunity through thymosin.
What is the key function of lymph nodes?	They filter lymph and destroy bacteria and toxic substances through macrophages and lymphocytes.

27. Extra Last-minute Points to Fill the Answer

When writing an answer on blood, do not stop after listing the components. Add why each component matters: plasma maintains volume and transports substances; RBCs transport respiratory gases; WBCs defend against infection; platelets prevent bleeding. This converts a simple list into a physiology answer.

For anemia, always connect the definition with oxygen transport. Low hemoglobin means low oxygen-carrying capacity. The body compensates by increasing heart rate and respiration, which explains palpitation and breathlessness. Add RBC morphology wherever possible because it helps identify the type of anemia.

For coagulation, draw the cascade only if labels are readable. Write the three-stage summary even when you draw: prothrombin activator formation, thrombin formation and fibrin formation. Mention calcium and vitamin K-dependent factors to show applied understanding.

For immunity, write in paired form: innate versus acquired, T cell versus B cell, cellular versus humoral, active versus passive. These pairs make the answer organized and reduce repetition. Always mention thymus with T cell maturation and thymosin.

Common mistake	Correct way to write
Writing plasma and serum as same.	Serum lacks fibrinogen and most clotting factors.
Writing WBC function only as defense.	Add phagocytosis, antibody formation, antigen presentation, allergy and inflammation.
Forgetting platelet role in repair.	Add PDGF helps repair of damaged blood vessels.
Writing ABO without antibodies.	Mention both antigen on RBC and antibody in plasma.
Writing lymph as only tissue fluid.	Add lymph returns proteins, absorbs fats and carries antigens

to lymph nodes.

Diagram to draw	Labels that must be visible
RBC	Biconcave shape, central pallor, diameter and rouleaux if asked.
WBC types	Nucleus shape, granules and name of each cell.
Coagulation cascade	Intrinsic, extrinsic, factor X, prothrombin, thrombin, fibrinogen and fibrin.
Blood grouping	Anti-A, anti-B, agglutination and final group.
Lymph node	Capsule, cortex, paracortex, medulla, afferent and efferent lymphatics.
Antibody	Heavy chain, light chain, variable region and antigen-binding sites.

Final check before exam: revise normal values, diagrams and classifications first. Most Unit II questions are scoring because the same facts can be reused across long answers, short notes and MCQs.

Source note: Content and diagrams are prepared from K. Sembulingam and Prema Sembulingam, *Essentials of Medical Physiology*, 6th edition, aligned with Unit II syllabus.